

ternating among GPT-4o-mini, GPT-4o, Llama3-8B, and Qwen2.5-7B during successive paraphrasing. Although each model brings its own stylistic biases, the fundamental attractor cycle remains intact. Interestingly, perplexity computed by a single model (e.g., Llama3-8B) on paraphrases generated by other models still decreases over iterations in Figure 6. This suggests that the attractor states are not confined to a single model’s parameter space. Instead, they reflect a more general statistical optimum that multiple LLMs gravitate toward.

From a systems perspective, this findings suggest that randomizing the transformation function P does not inherently break the attractor. The system remains in a basin of attraction shared across these varied modeling conditions, implying that the stable cycle is a robust property of the iterative transformation rather than a quirk of any particular prompt or model.

5.3 Increasing Generation Randomness

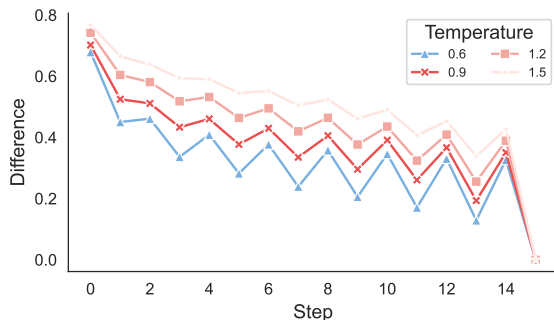


Figure 7: The difference between T_{15} and T_i generated by GPT-4o-mini. By increasing the temperature, randomness is amplified, causing the differences to grow as well.

Another strategy is introducing more stochasticity in the generation process by increasing the generation temperature. Higher temperatures expand the immediate token selection space, potentially allowing trajectories to wander away from the attractor. However, as shown in Figure 7, while higher temperatures do increase the difference between successive paraphrases, the system still exhibits a 2-period cycle. Further increases in temperature lead only to nonsensical outputs. This outcome aligns with dynamical systems theory: a small increase in stochasticity may create local perturbations, but if the basin of attraction is strong, the system remains near the limit cycle. Excessive stochastic forcing can push the system out of meaningful regions of

state space entirely, leading to “chaotic” or nonsensical behavior, rather than discovering a new stable attractor with richer linguistic diversity.

5.4 Experiments with Complex Prompts

Previous experiments were conducted using a simple paraphrase prompt, leading to existing limitations. To solve this, we experimented with a more complicated prompt, and the results indicated similar periodicity patterns. This prompt forces LLMs to enhance grammatical and syntactical variety. We used this prompt to instruct GPT-4o-mini to successively paraphrase the paragraph-level test set for 15 rounds. The empirical evaluation of periodicity (2-periodicity score) and convergence (PPL) of the successive paraphrasing with the complex prompt is listed below. Both the difference confusion matrix and the prompt are shown in Appendix C.3.

Model	Periodicity	Convergence
Original	0.80	1.19
Complex	0.67	1.33

Table 3: Periodicity and Convergence Table

Although the sophisticated prompt alleviated the periodicity and convergence in some degree, the pattern of 2-period cycle remained strong. For context, a periodicity score of 0.67 implies an average edit distance of 0.33 between paraphrases two steps apart, whereas direct paraphrase exhibits an edit distance of 0.68.

5.5 Incorporating Local Perturbations

We introduce local perturbations to mitigate the attractor cycle pattern. At the end of each iteration, we edit 5% of the text by introducing perturbations using three methods: synonym replacement (S.R.), word swapping (W.S.), and random insertion or deletion (I/D.). As shown in Table 4, among these interventions, synonym replacement barely affects periodicity, suggesting that minor lexical changes do not move the system out of the attractor’s basin. It indicates that except during the first paraphrasing, LLMs primarily perform synonym replacements for words or phrases, as shown in Figure 1. Word swapping, however, causes more significant disruption, lowering periodicity more effectively. From a dynamical standpoint, large structural perturbations are needed to shift the system’s state out of a stable cycle. Local lexical tweaks do not suffice because the attractor’s pull is strong and preserved at a deeper structural level.

w/o Perturb.	S.R.	W.S.	I/D.
0.77	0.73	0.62	0.66

Table 4: Impact of different types of perturbations on 2-periodicity degrees τ , compared to the original text during paraphrasing.

5.6 Paraphrasing with History Paraphrases

We consider a scenario where the transformation $\hat{P}$ depends on both T_i and T_{i-1} . This added historical context can alter the equilibrium states. In a scenario where we paraphrase T_i based on the reference T_{i-1} , it is essential that T_{i+1} differs from both T_i and T_{i-1} . This function can be expressed as: $T_{i+1} = \hat{P}(T_i, T_{i-1})$. In this context, P_{i-1} emerges as a strong candidate for paraphrasing $P(T_{i+1}, T_i)$, as it aligns with the distribution of LLMs while maintaining difference from $\hat{P}(T_{i+1}, T_i)$, satisfying the task requirement. As a result, this more complex cycle still represents a stable attractor, albeit of higher order, as shown in Figure 8.

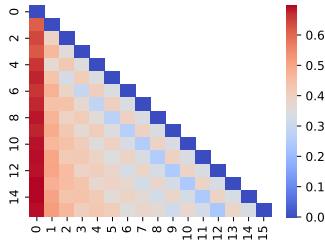


Figure 8: When adding historical paraphrases, LLMs exhibit 3-periodicity in the paraphrasing task.

5.7 Sample Selection Strategies

We investigate methods to steer the system away from stable attractors at the least cost of generation quality. Given the correlation between periodicity and perplexity, it is intuitive to mitigate this issue by increasing perplexity while maintaining generation quality. To achieve this, we can randomly sample multiple paraphrases at each iteration and select the one based on perplexity. We design three types of strategies: selecting the paraphrase with the maximum or minimum perplexity or randomly choosing one at each iteration. Figures 9 illustrate that selecting a higher perplexity can reduce periodicity. However, such diversity comes at the cost of semantic equivalence (Appendix C.5). Considering both periodicity and meaning preservation, we recommend the random strategy, which effectively

reduces periodicity while incurring minimal information loss compared to selecting the option with the lowest perplexity.

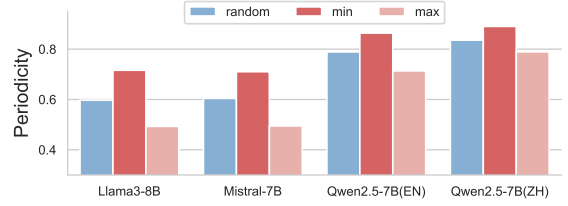


Figure 9: The periodicity of three strategies using different LLMs.

5.8 The Benefit of Mitigating Small-Size Cycles

As a data augmentation method, paraphrase diversity should impact downstream tasks. To figure out this, we conducted an experiment on domain classification using successive paraphrasing for data augmentation. We selected a commonly used dataset, AG News (Zhang et al., 2016) as the testbed, and trained BERT-based models using different data. For data augmentation, we conducted 5 rounds of successive paraphrasing under two different settings—min-strategy (Min. Strat.) and max-strategy (Max. Strat.), as detailed in Section 5.7. Below are our results:

Metric	w/o Aug.	Min. Strat.	Max. Strat.
Accuracy $\uparrow$	83.10%	83.80%	84.41%
2-periodicity $\downarrow$	-	0.51	0.33

Table 5: Performance on AG News across different data augmentation strategies.

The table shows that our max-strategy, which more effectively mitigates the 2-periodicity cycle, yields more diverse paraphrases for data augmentation and therefore achieves higher classification accuracy (84.41%) compared to both no augmentation and the min-strategy.

6 Related Work

Paraphrase Generation. Paraphrase generation has long been a significant focus in NLP research, with numerous studies dedicated to enhancing the quality of generated paraphrases. (Li et al., 2018; Roy and Grangier, 2019; Lewis et al., 2020; Lin et al., 2021b; Hosking et al., 2022; Xie et al., 2022). Some studies also explore methods to control paraphrase generation by focusing on aspects such as

syntactic and lexical diversity (Li et al., 2019; Goyal and Durrett, 2020; Huang and Chang, 2021; Bandel et al., 2022; Krishna et al., 2023; Yang et al., 2022). Others investigate the application of paraphrase generation as a data augmentation technique to enhance model performance (Jolly et al., 2020; Bencke and Moreira, 2024; Okur et al., 2022b).

Recently, advancements in LLMs have enabled LLM-based paraphrasing tools to generate stable, high-quality responses, making them widely used for refining materials like news articles, academic papers, and speeches (Witteveen and Andrews, 2019; Roe and Perkins, 2022b; Rani et al., 2023). However, their work primarily discusses single-step paraphrasing. In contrast, another line of work involves LLMs iteratively rephrasing their own outputs over multiple iterations. Sadasivan et al. (2023) explores how repeated rephrasing can help evade AI text detectors, while Tripto et al. (2023) and Huang et al. (2024) discuss the implications for authorship after a document has undergone multiple rounds of paraphrasing. Our research differs from those works. We investigate the inherent characteristics of paraphrasing when extended over multiple iterations.

Self-Reinforcement in LLMs. Repetition, defined as the occurrence of repetitive text in natural language generation, has been widely explored in research community (Holtzman et al., 2020a; Welleck et al., 2020; Lin et al., 2021a; See et al., 2017; Liu and Lapata, 2019; Fu et al., 2021). Several studies have observed repetition in text generation (Holtzman et al., 2020b; Finlayson et al., 2023) and proposed various sampling strategies to mitigate this issue. Ivgi et al. (2025) explores the connection between repetition and model size. Additionally, Xu et al. (2022) introduce the concept of self-reinforcement to elucidate this phenomenon, demonstrating that LLMs exhibit a tendency to repeat preceding sentences and reinforce this behavior during generation. Yan et al. (2024) further explore the relationship between the self-reinforcement effect and the in-context learning capabilities of LLMs. While the repetition explored in these studies primarily focuses on single-round generation scenarios, our research reveals a similar phenomenon in multi-round generation. We specifically examine the self-reinforcement patterns of LLMs across successive paraphrasing tasks and concentrate on typical behaviors observed in them.

7 Conclusion

We reframed successive paraphrasing as a discrete dynamical system, offering a principled explanation for the emergence of stable periodic attractors in LLM-generated text. Our empirical findings revealed that instead of producing an expanding array of diverse paraphrases, LLMs rapidly settled into low-order limit cycles. These attractor states persisted even when we vary models, prompts, generation temperatures, and local kick perturbations, indicating that they stem from a fundamental property of the system rather than superficial repetition or particular model idiosyncrasies. Viewing iterative text generation through the lens of systems theory helps clarify why certain interventions fail to break these cycles and how others can weaken the attractor’s pull. Ultimately, recognizing and addressing these stable attractor cycles is crucial for unlocking more expressive and flexible language generation for large language models.

Limitations

While this study provides valuable insights into successive paraphrasing, several limitations should be acknowledged. First, the paraphrasing is based on simple prompts, which may limit the generalizability of the findings to more complex or specific prompts. Second, although we have examined this phenomenon in the currently prevalent LLMs, other LLMs may not exhibit the same behavior. Finally, while we present the convergence of reverse perplexity in this work, the underlying reasons for this behavior still require further investigation.

Ethic Considerations

We uphold the Code of Ethics and ensure that no private or non-public information is used in this work. We comply with the terms set by companies offering commercial LLM APIs and extend our gratitude to all collaborators for their invaluable support in utilizing these APIs.

Acknowledgement

We extend our gratitude to all the reviewers for their valuable feedback and suggestions, which greatly contributed to enhancing the quality of the paper. This research was supported by the Natural Science Foundation of China Key Program under Grant Number 2022YFE0204900.